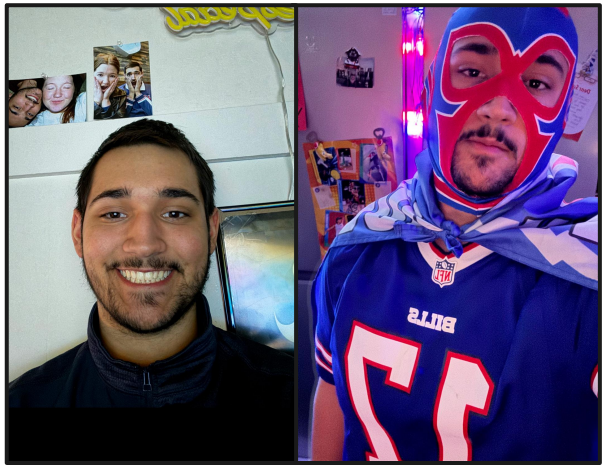
Mobile Forensics in Law Enforcement

By: Max Carney & Noah Cacio

Who are we?

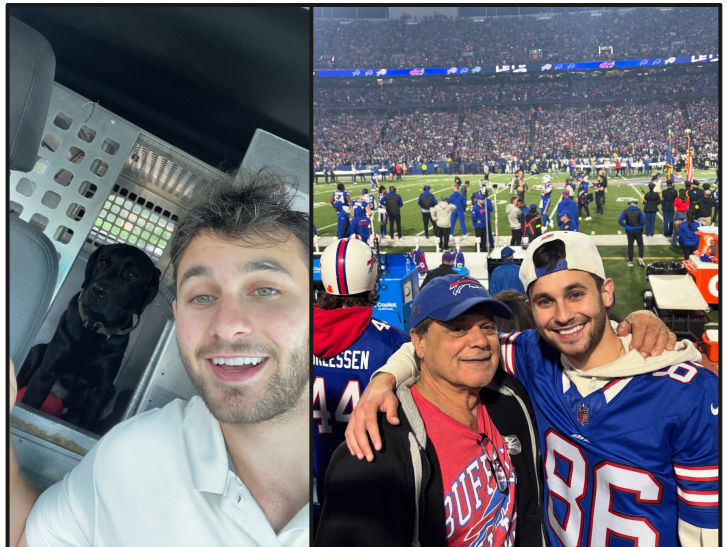
Max Carney

- Computer & Digital Forensic major
- Worked at the Leahy Center for Digital Forensics & Cybersecurity for 3 years
- Current intern at Envista Forensics
- Huge Buffalo Bills fan



Noah Cacio

- Computer & Digital Forensic major
- Interned with Lincoln NH. PD/NH ICAC for last 4 years
- Passionate about ICAC's mission
- Will be employed at Lincoln NH. PD post graduation
- Huge Buffalo Bills fan

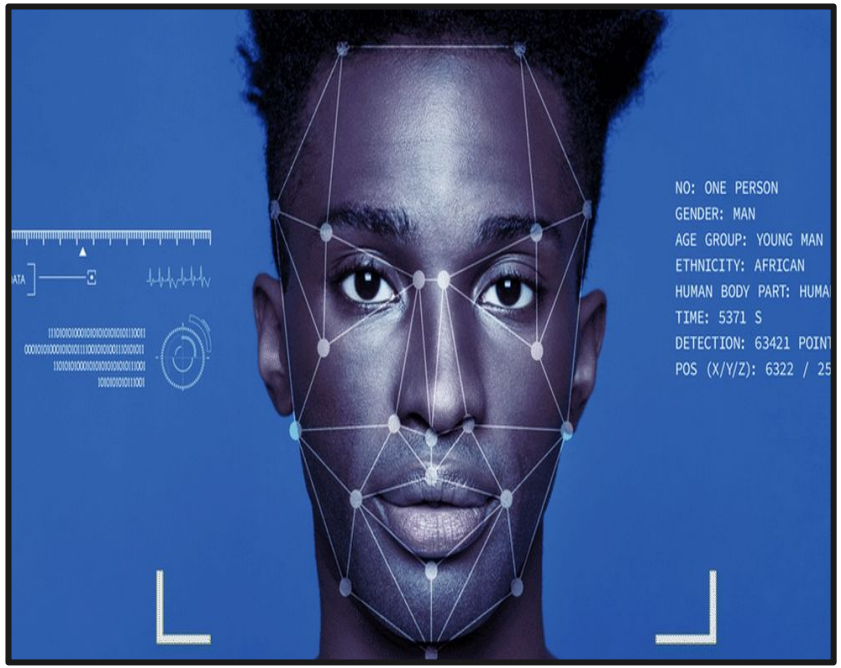


What is Mobile Forensics?

- The extraction and analysis of data from smartphones, tablets, and other mobile devices
- Recovers deleted content including texts, call logs, photos, emails, and app data
- Bypasses screen locks and encryption using specialized forensic tools like Cellebrite or Oxygen Forensics
- Tracks location history through GPS data, cell tower logs, and Wi-Fi connection records
- Critical in modern investigations as mobile devices often contain the most personal and incriminating digital evidence

What information can be found?

- Personal information
- Social standing within the community as well as with family, friends, and potentially the victim
- Other incriminating information that otherwise would have not been located
- File sharing
- Location data
- Transaction histories (In person or online shopping)
- Browsing information (Jose Pazos case)

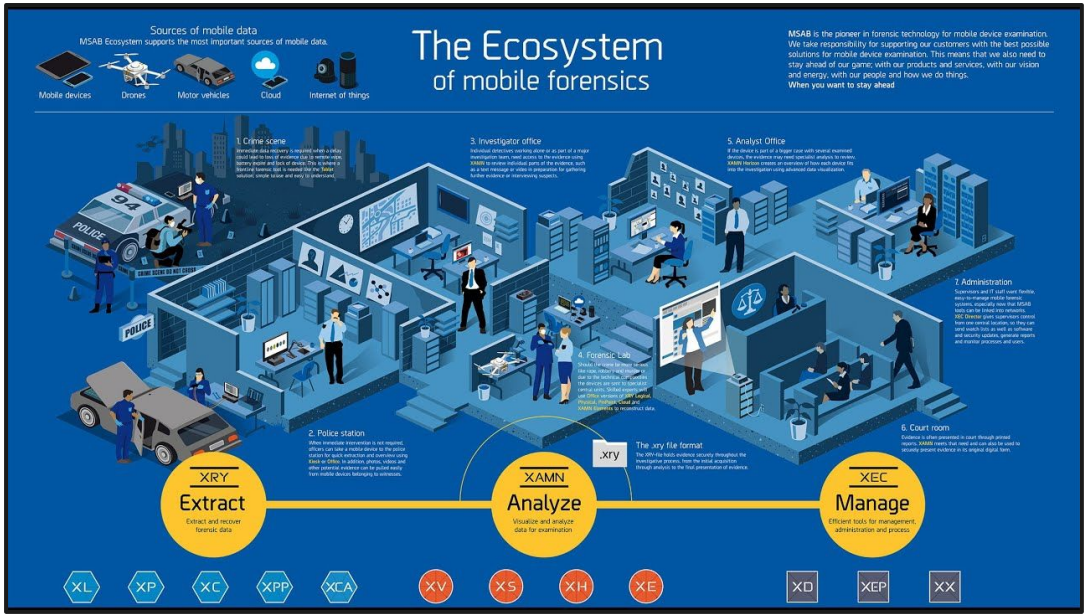


How is this relevant to legal proceedings?

- One message can be used to obtain a warrant for arrest
- In the case of illegal content, each file is its own charge
- Digital evidence can serve as the primary evidence in a case
- Can create a clear timeline of the case
- Identifies any collaborators
- In certain instances help establish a motive
- Supports or contradicts certain alibis
- Concrete evidence so it enforces accountability and ensures authenticity

What is the process?

- Gathering evidence
- Imaging
- Investigating
- Documenting



Tool 1: Axiom

- A comprehensive forensics platform by Magnet Forensics for acquiring evidence from mobile, cloud, and computer sources
- Recovers artifacts from 1,000+ sources including iOS, Android, Windows, macOS, and major cloud services
- Uses AI-powered analysis to automatically categorize evidence and surface the most relevant data faster
- Displays a simple timeline view allowing investigators to reconstruct events across multiple devices

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Where is Axiom used?

- Law enforcement agencies use it to investigate crimes involving digital evidence like fraud, trafficking, and cybercrime
- Corporate security teams deploy it for internal investigations and employee misconduct cases
- Government and intelligence agencies rely on it for national security and counterterrorism investigations
- Legal and litigation teams use it to gather court-admissible digital evidence for civil and criminal cases
- Incident response teams utilize it to investigate data breaches and network compromises

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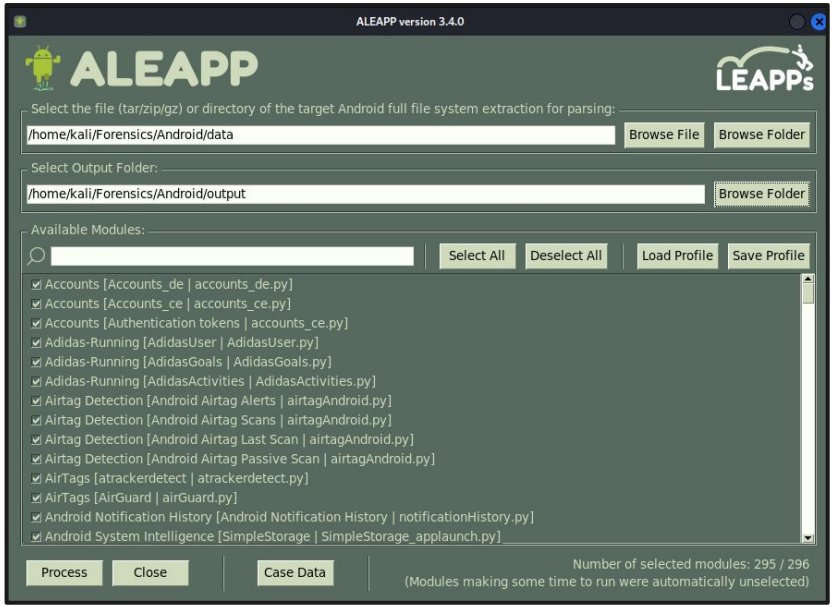
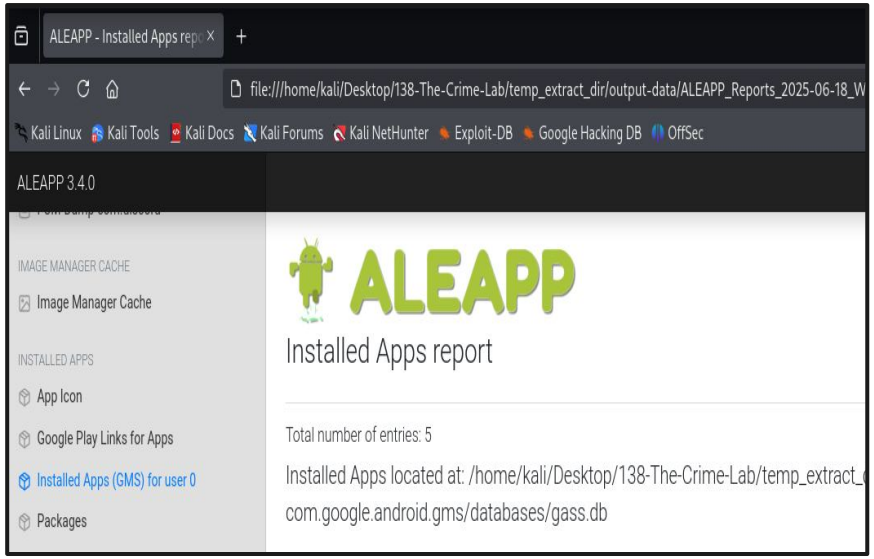
Tool 2: ALEAPP

- A free, open-source tool designed specifically for parsing and analyzing Android device artifacts
- Processes hundreds of Android data sources including call logs, contacts, app data, and location history
- Extracts data from app databases such as WhatsApp, Instagram, Snapchat, and other popular applications
- Generates easy to read HTML reports allowing investigators to quickly review and present findings
- Widely used by forensic examiners as a fast and lightweight alternative to expensive commercial tools



Where is ALEAPP Used?

- Law enforcement investigators use it to quickly parse Android devices during criminal investigations
- Digital forensics students and researchers rely on it as a free learning tool for mobile artifact analysis
- Corporate investigators deploy it for employee misconduct cases involving Android devices
- Incident response teams use it to rapidly triage Android devices during security incidents
- Small agencies and nonprofits utilize it as a cost-effective alternative when budget limits access to commercial tools



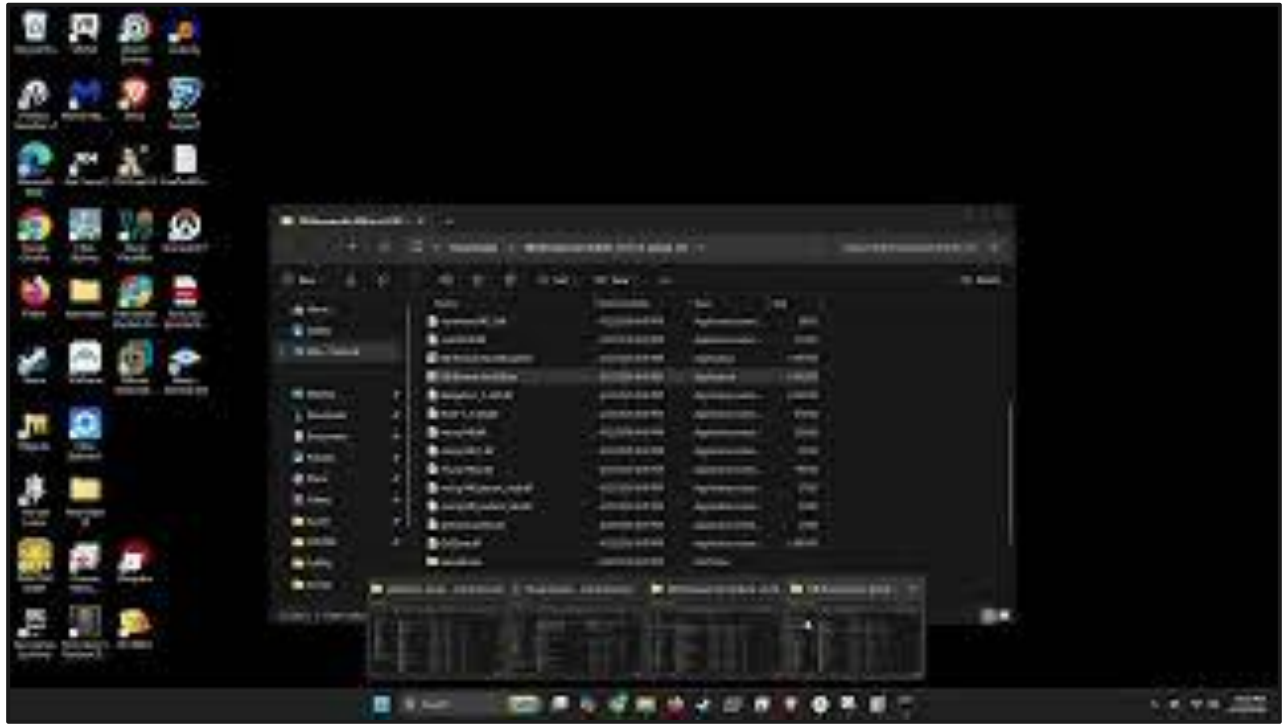
What about unprocessed information?

- ALEAPP can miss artifacts stored in obscure or app-specific SQLite databases not yet supported by the parser
- Third party app databases like dating apps and lesser-known messengers often go unprocessed
- Deleted SQLite records may still exist in unallocated space and require manual recovery
- WAL log files attached to SQLite databases can contain recent activity that tools frequently overlook
- DB Browser allows investigators to manually query raw database tables and uncover missed evidence

DB Browser for SQLite

- A free, open-source database viewer that allows users to create, view, and edit SQLite database files
- Provides a simple visual interface to browse tables, records, and fields without needing SQL knowledge
- Allows execution of custom SQL queries to search and filter specific data within a database
- Commonly used in forensics to manually examine app databases extracted from mobile devices
- Supports import and export of data in formats like CSV and JSON for further analysis or reporting

Example video of DB Browser



DB Browser Database Parsing

DB Browser for SQLite - C:\Users\maxca\AppData\Local\Android\Sdk\platform-tools\com.instagram.android\databases\direct.db

File Edit View Tools Help

New Database | Open Database | Write Changes | Revert Changes | Undo | Open Project | Save Project | Attach Database | Close Database

Database Structure | Browse Data | Edit Pragmas | Execute SQL

Table: messages | Filter in any column

	cipient_ids	timestamp	message_type	text
		Filter	Filter	Filter
1	37713421	1776038451127346	text	You are too cute buddy. Nobody is scared of you and your garbage takes, I promise!
2	37713421	1776038587807545	text	I am cute? How about you find out how cute Dan Stone gets. It ain't Dan Flowers for a reas
3	37713421	1776038741232377	text	Time and place?
4	37713421	1776038787999479	text	Tommorrow, Dicks Sporting Goods parking lot, 9pm. Be there or be square. Dorktimusprime
5	37713421	1776038827274727	text	Lmao ur too funny. See you there buddy
6	37713421	1776127850101155	text	I'll be awaiting you...

What do we get for this? (what deliverables)

- Cellphone Warrants
- Subpoenas
- Arrest Warrants
- Chain of custody forms
- Media release forms

Questions?

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